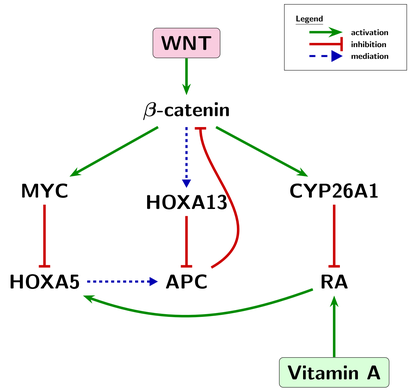
Physics-Informed Neural Networks for Forward Modeling and Parameter Inference in a Reduced WNT-RA-HOX Model of Colorectal Cancer Initiation

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01 Model & formulation

From signaling to stemness. Seven ODEs connect WNT, APC and retinoic-acid regulation. We study dynamics, parameter recovery and uncertainty.



Disease regimes

Regime	W	θ_P
Normal	0.8	1.00
Early	1.0	0.75
Advanced	1.5	0.50
Severe	2.0	0.25

W: WNT drive.
θ_P: APC functioning.

Seven states

$\mathbf{z} = (b, p, h_5, h_{13}, m, r, c)$
b: β-catenin; p: APC
h_5, h_{13}: HOXA5, HOXA13
m: MYC; r: RA; c: CYP26A1

Fig. 1. Regulatory network. Green: activation; red: inhibition; blue dashed: mediation. Early and Advanced denote adenoma.

Nondimensionalization

Scale concentrations by characteristic values and time by β-catenin turnover, $d_B = 1\,\mathrm{h}^{-1}$:

$$X = X_* x, \qquad \tau = d_B t, \qquad \varepsilon_X = \frac{d_B}{d_X}. \tag{1}$$

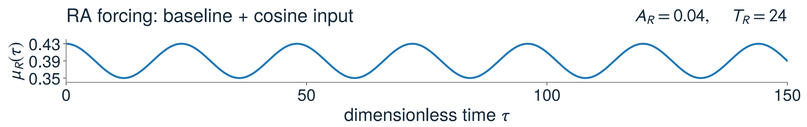
Hill activation: $H_n(u;\kappa) = u^n/(\kappa^n + u^n)$. A dot below denotes $d/d\tau$.

$$\begin{aligned}
\dot{b} &= W + \eta_{13} H_{n_H}(h_{13};\kappa_{13}) - b - \lambda_P p b - \lambda_5 \frac{h_5 b}{\kappa_5 + b},\\
\varepsilon_P \dot{p} &= \frac{1+\rho_5 h_5}{1+\rho_B b + \rho_{13} h_{13}} - D_P(\theta_P) p,\\
\varepsilon_5 \dot{h}_5 &= a_5 + \eta_R H_1(r;\kappa_R) - h_5 - \eta_M \frac{m h_5}{\kappa_M + m},\\
\varepsilon_{13} \dot{h}_{13} &= a_{13} + \eta_{B13} H_{n_B}(b;\kappa_{B13}) + \eta_{M13} H_{n_M}(m;\kappa_{M13}) - h_{13},\\
\varepsilon_M \dot{m} &= a_M + \eta_{BM} H_{n_B}(b;\kappa_{BM}) - m,\\
\varepsilon_R \dot{r} &= \mu_R(\tau) - r - \lambda_C c r,\\
\varepsilon_C \dot{c} &= a_C + \eta_{RC} H_1(r;\kappa_{RC}) + \eta_{BC} H_{n_B}(b;\kappa_{BC}) - c.
\end{aligned} \tag{2}$$

APC functioning and retinoid forcing

$$\begin{aligned}
D_P(\theta_P) &= 1 + \delta_{P1}(1-\theta_P),\\
\mu_R(\tau) &= \mu_0 + A_R\left[1 + \cos\left(\frac{2\pi\tau}{T_R} - \varphi\right)\right]\\
&\quad + \frac{D_R}{2}\left[\tanh q(\tau-\tau_1) - \tanh q(\tau-\tau_2)\right].
\end{aligned} \tag{3}$$

Reference: Radau on $\tau \in [0, 150]$; ATRA pulse $[40, 88]$. Relative timescales $\varepsilon_R = 0.40$, $\varepsilon_M = 0.60$ produce mild stiffness.



02 Dynamics & PINN reconstruction

Normal regime: $W = 0.8$, $\theta_P = 1$; MYC and APC share each axis.

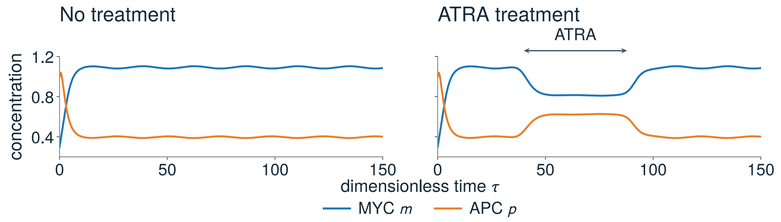
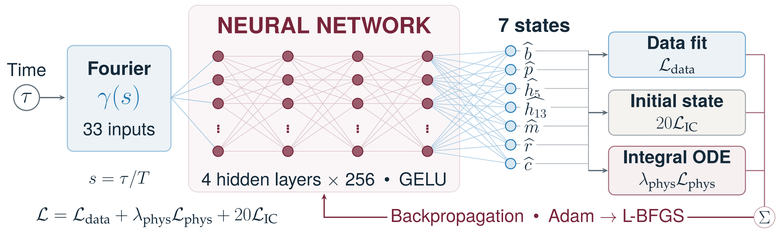


Fig. 2. Radau solutions; baseline circadian forcing $A_R = 0.04$ in both panels.



$$\mathbf{r}_k = \widehat{\mathbf{z}}_{k+1} - \widehat{\mathbf{z}}_k - \frac{\Delta\tau_k}{2}\left(\mathbf{F}_k + \mathbf{F}_{k+1}\right). \tag{4}$$

Ten networks share 36 parameters; 150 times/condition ($\sigma = 0.002$).
Relative state weights; adaptive λ_{phys}; two starts.[1,2]
IC weight 20: heuristic emphasis on the known initial state.

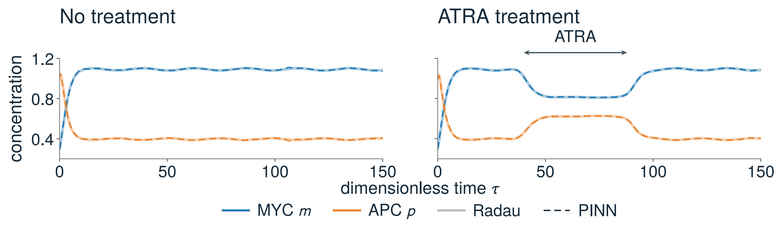


Fig. 3. Saved inverse-PINN fits (dashed) and Radau (solid). Relative L^2 errors, MYC/APC: untreated 0.31 /0.75 %; treated 0.40 /0.54 %.

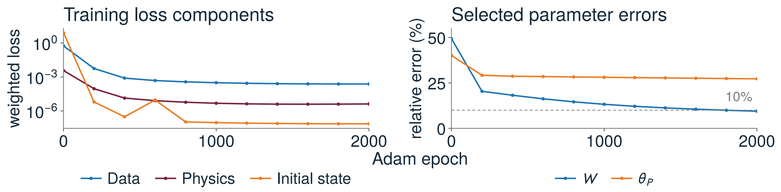


Fig. 4. Saved Adam checkpoints from the selected Normal run; losses average ten conditions. Final fits also include L-BFGS.

Parameters within 10% of truth

Method	Normal	Early	Advanced	Severe	Total
Autodiff	16	9	6	6	37
Integral	17	16	10	7	50

Table 1. Ten conditions per method: 37 → 50 of 144 parameter–regime pairs. The gain belongs to the integral implementation as a whole.

03 Identifiability & Bayesian uncertainty

Fisher information and parameter recovery

Local sensitivities at truth: 7 states, 10 conditions and 200 times.

$$\mathcal{I} = \sigma^{-2} J^{\mathsf{T}} J, \qquad J_{ki} = \frac{\partial y_k}{\partial \log \theta_i}, \qquad \sigma = 0.002. \tag{5}$$

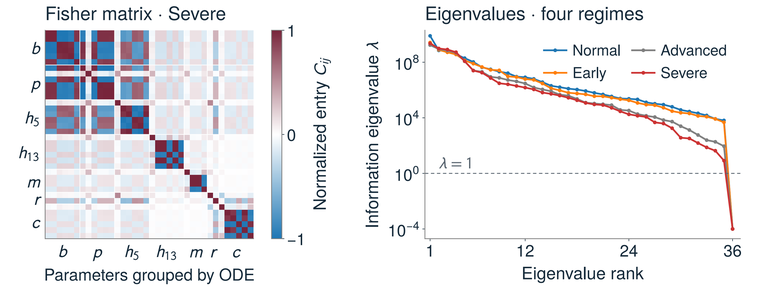


Fig. 5. Severe-regime $C_{ij} = \mathcal{I}_{ij}/\sqrt{\mathcal{I}_{ii}\mathcal{I}_{jj}}$: $|C_{ij}| \approx 1$ signals overlapping sensitivities. Each regime has **one near-null direction** ($\lambda < 1$); plotting floor 10^{-4}.

Bayesian parameter distributions

Frozen-state HMC samples all 36 parameters using the integral likelihood.[3]

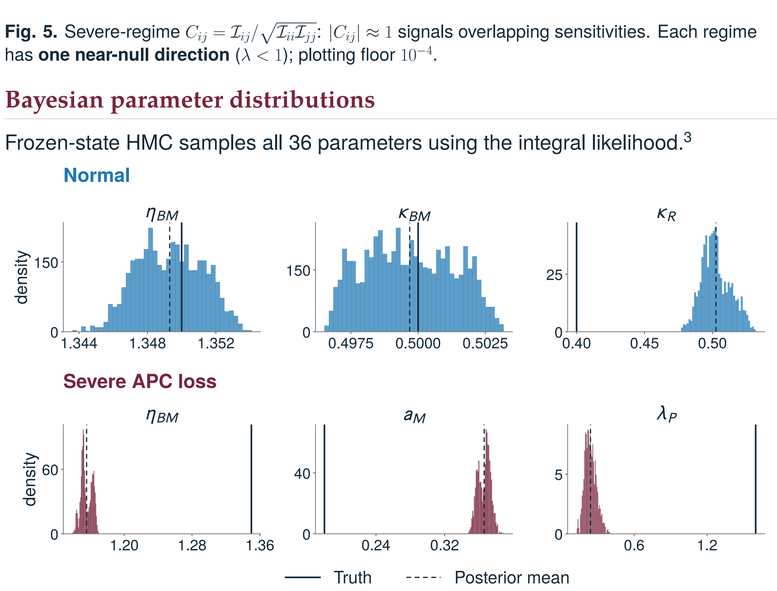


Fig. 6. Presentation marginals (2000 draws/regime): narrow distributions can miss truth.

Calibration limit. HMC fails effective-sample-size and coverage checks; interval widths are unreliable.

Fisher analysis of eight parameters

We varied eight parameters and fixed the other 28 at their true values:

$$\lambda_C,\ \eta_{BM},\ \kappa_{BM},\ \kappa_{BC},\ \eta_{BC},\ \eta_{RC},\ \kappa_{B13},\ \kappa_R$$

Fisher result	Normal	Early	Advanced	Severe
Locally identifiable	8/8	8/8	8/8	8/8
Eigenvalues below 1	0	0	0	0
$\mathrm{cond}(\mathcal{I})$	99.7	120.9	229.4	665.2

Table 2. The observations match the full Fisher analysis above. PINN recovery of these eight parameters has not been tested.

Scope and next step

Synthetic-data study. Treatment changes only D_R; all other forcing is retained. Fisher diagnostics are local; HMC intervals are uncalibrated. Next: reduced-parameter PINN fits and validation across noise and sampling levels.

Key references

[1] M. Raissi, P. Perdikaris and G. E. Karniadakis. Physics-informed neural networks. *J. Comput. Phys.* 378, 686–707 (2019). doi:10.1016/j.jcp.2018.10.045.
[2] A. Yazdani et al. Systems biology informed deep learning. *PLoS Comput. Biol.* 16, e1007575 (2020). doi:10.1371/journal.pcbi.1007575.
[3] L. Yang, X. Meng and G. E. Karniadakis. B-PINNs. *J. Comput. Phys.* 425, 109913 (2021). doi:10.1016/j.jcp.2020.109913.